Highlights

Bitcoin Direction Prediction via Dual-Encoder Cross-Attention: Interpretability and Uncertainty-Aware Trading

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- Walk-forward cross-validation (CV) exposes optimistic bias in simple train-test splits, reversing CNN-LSTM (Convolutional Neural Network – Long Short-Term Memory) performance.
- The proposed DECA (Dual-Encoder Cross-Attention) architecture achieves the highest Matthews Correlation Coefficient (MCC = 0.648) among all evaluated models.
- On-chain blockchain metrics provide approximately ten times more predictive signal than technical indicators alone.
- Monte Carlo (MC) Dropout raises prediction accuracy to 96.4% at 50% coverage via calibrated uncertainty filtering.
- A dynamic MC confidence threshold achieves 4,013%/year annualised return, surpassing deterministic DECA (3,922%) in backtesting.

Bitcoin Direction Prediction via Dual-Encoder Cross-Attention: Interpretability and Uncertainty-Aware Trading

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Abstract

This study investigates the performance of Machine Learning (ML) and Deep Learning (DL) models in predicting the next-day direction of Bitcoin price returns, with direct application to long/short algorithmic trading strategies. The model inputs comprise Bitcoin on-chain blockchain metrics and classical technical analysis indicators. The ML and DL models evaluated include Support Vector Machines (SVM), XGBoost, LightGBM, Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM), and a Multilayer Perceptron (MLP). A central contribution of this study is the proposed Dual-Encoder Cross-Attention (DECA) architecture, which routes each input type through independent residual encoders and integrates their cross-domain interactions via cross-attention, providing interpretable attention weights and uncertainty-aware predictions through Monte Carlo Dropout. An important methodological aspect is the adoption of an expanding-window walk-forward cross-validation protocol covering six annual folds from 2019 to 2024, which offers more rigorous temporal validation than the simple train-test splits commonly adopted in the literature. The findings reveal that DECA achieves the highest Matthews Correlation Coefficient (MCC = 0.648) among all models evaluated, while remaining statistically equivalent to SVM in directional accuracy. Static models generally outperformed sequence models under this evaluation protocol. It was found that on-chain blockchain metrics provide substantially greater directional signal than technical indicators alone, corroborating and extending recent findings in the literature. Finally, the study assessed model profitability through backtesting by replicating the conditions of prior work, and DECA emerged as the most profitable deterministic model. A dynamic confidence threshold on the Monte Carlo uncertainty further improved returns. This study emphasizes the importance of evaluation methodology, architecture design, and input data type in Bitcoin price direction prediction. The findings offer practitioners principled tools for uncertainty-aware trading decisions and improved investment outcomes.

Keywords: Bitcoin, price direction prediction, dual-encoder cross-attention, on-chain metrics, walk-forward cross-validation, Monte Carlo Dropout, uncertainty quantification

1. Introduction

Bitcoin remains the world's largest cryptocurrency by market capitalisation (CoinMarketCap, 2024), with daily returns frequently exceeding ±5% and structural regime changes driven by halving cycles, institutional adoption, and regulatory developments (Baur et al., 2018; Dyhrberg, 2016). Predicting the *direction* of next-day returns (binary classification: up or down) is more directly actionable than point-value regression, as it maps cleanly to long/short trading decisions while remaining less sensitive to magnitude errors.

A substantial body of research has addressed this problem over the past decade, achieving self-reported accuracies in the range of 75–83% (McNally et al., 2018; Omole and Enke, 2024, 2025; Dubey and Enke, 2025). However, the majority of these results are obtained on a single 80/20 train-test split, a protocol known to produce optimistically biased estimates when applied to time-series data (Bergmeir and Benítez, 2012; Tashman, 2000). As we show in

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Section 5, results for sequence-based models differ substantially across evaluation protocols, and the gap between simple-split and walk-forward estimates can be considerable, consistent with the time-series CV literature (Bergmeir and Benítez, 2012; Tashman, 2000).

A second limitation of prior work is the separation between technical and on-chain features. Omole and Enke (2024) used only on-chain metrics; Omole and Enke (2025) subsequently combined both sources but processed them as a single flat feature vector. Neither study exploited the structural complementarity between these two domains through an architecture designed to model their interaction explicitly: technical indicators summarise price dynamics, while on-chain metrics encode blockchain-level investor behaviour.

In this paper we address both limitations with four contributions:

1. Dual-Encoder Cross-Attention (DECA) architecture. We introduce DECA, a model that routes technical indicators and on-chain metrics through domain-specific deep MLP encoders, then learns cross-domain interactions via bidirectional cross-attention (Vaswani et al., 2017) with learnable context tokens and gated fusion. Under walk-forward CV, DECA achieves MCC = 0.648, the highest across all five models evaluated. DECA and SVM (MCC = 0.633) form a statistically equivalent top tier (McNemar $p = 1.0$); their differentiation lies in operational capabilities: DECA's cross-attention weights provide domain-level interpretability and MC Dropout yields calibrated uncertainty estimates unavailable to kernel methods.
2. Rigorous walk-forward evaluation with Bayesian hyperparameter optimisation. We apply a six-fold expanding-window cross-validation protocol covering 2019–2024, with Optuna TPE optimisation (Akiba et al., 2019) running 50 trials per model independently on each architecture. Statistical significance is assessed via McNemar's test (Dietterich, 1998) with Bonferroni correction over 10 pairwise comparisons ($\alpha = 0.005$). This methodology directly addresses the evaluation gap identified in prior work (Bergmeir and Benítez, 2012).
3. MC Dropout uncertainty quantification for actionable trading. We apply Monte Carlo Dropout (Gal and Ghahramani, 2016) to DECA at no retraining cost (30 stochastic passes), obtaining well-calibrated epistemic uncertainty. Filtering to the 50% most confident predictions raises accuracy to 96.4% (2019–2024), enabling principled coverage–accuracy trade-offs in practical trading strategies.
4. Ablation study over four feature groups. We quantify the individual contribution of OHLCV, technical indicators, and on-chain metrics via XGBoost ablation. On-chain metrics alone (G2) outperform technical indicators alone (G1) by a factor of approximately ten in MCC (0.387 vs. 0.036), corroborating and extending the findings of Omole and Enke (2025).

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset and feature engineering. Section 4 details the proposed architecture, baselines, and evaluation protocol. Section 5 presents quantitative results and analysis. Section 6 concludes.

2. Related Work

2.1. Bitcoin Direction Prediction

Early machine learning approaches to Bitcoin price prediction followed stock-market paradigms, applying SVMs, random forests, and LSTMs to price-derived technical indicators (McNally et al., 2018; Patel et al., 2015). A common finding is that no single architecture consistently dominates across evaluation windows (Ji et al., 2019), and that results are heavily dependent on the choice of train-test split.

Subsequent work incorporated alternative data sources, most notably social media sentiment. Zhang et al. (2011) demonstrated that Twitter sentiment polarity provides a statistically significant leading signal for next-day Bitcoin returns, and Abraham et al. (2018) showed that Reddit discussion volume correlates with short-term volatility spikes. Hybrid models that concatenate sentiment embeddings with price-derived features have been shown to improve directional accuracy over price-only baselines (Valencia et al., 2019), though the gains tend to degrade during periods of abnormally high posting volume such as market crashes.

Several studies have designed architectures specifically for cryptocurrency rather than adapting stock-market models. Lahmiri and Bekiros (2019) showed that LSTM networks trained on hourly crypto data exhibit markedly different loss landscapes than equivalent models on equity indices, attributing this to the fat-tailed return distribution and the absence of circuit breakers in 24/7 markets. The problem of non-stationarity is particularly acute: Bitcoin has

transitioned through at least three structurally distinct regimes (retail speculation before 2017, institutional entry in 2020–2022, and spot-ETF-driven adoption from 2024 onward), and a model calibrated in one regime may be effectively out-of-distribution in the next (Bouri et al., 2019). Empirical evidence from this and related studies suggests that model performance can collapse abruptly when the underlying market structure shifts, consistent with a structural break rather than a recoverable noise floor.

Omole and Enke (2024) conducted a comprehensive study using 87 on-chain features and deep learning models (CNN-LSTM, LSTNet, TCN) with Boruta feature selection. With a CNN-LSTM on Boruta-selected features, they reported Accuracy = 82.4%, MCC = 0.649, AUC = 0.824 on a single 80/20 split of weekly Bitcoin data and demonstrated profitable backtesting results (6,653%/year annualised return).

Omole and Enke (2025) subsequently combined on-chain data with technical indicators (225 features total, Boruta selection) and found that SVM achieved the best accuracy (83%) on a simple holdout, while deep learning models (CNN-LSTM, TCN) performed comparably or worse. Crucially, technical indicators alone yielded only 50–53% accuracy (near-random), corroborating the finding replicated by our G1 ablation. More recently, Dubey and Enke (2025) revisited the same on-chain prediction task with Boruta, L1, and PCA feature selection paired with CNN-LSTM and TCN models, reporting 82.03% accuracy with Boruta + CNN-LSTM on a single holdout split and an annualised return of 1,682.7%. Our work extends these studies by (a) evaluating under walk-forward CV rather than simple splits, (b) introducing an architecture that explicitly models the interaction between technical and on-chain domains, and (c) providing uncertainty quantification applicable to risk-controlled trading strategies.

2.2. On-Chain Metrics as Predictive Signals

Blockchain transactions produce a publicly auditable trail of economic activity that has no direct analogue in traditional asset markets. Unlike equities, where fundamental data is disclosed quarterly and subject to accounting discretion, Bitcoin's ledger is continuously observable at transaction granularity.

Derived metrics synthesise this raw activity into economically interpretable signals. MVRV (Market Value to Realised Value) compares market capitalisation with the aggregate cost basis of all circulating coins; values below 1.0 historically correspond to accumulation phases that have preceded major bull markets (Puell and Mahmudov, 2018). SOPR (Spent Output Profit Ratio) measures the ratio of the USD value of coins at time of spending to their value at acquisition; sustained values below 1.0 signal that holders are capitulating at a loss (Shirakashi, 2019). Short-Term Holder SOPR (STH-SOPR) isolates coins moved within the past 155 days and functions as a real-time market stress indicator: sharp drops below 1.0 often mark local price floors. NVT (Network Value to Transactions) is a valuation ratio analogous to price-to-earnings; high NVT values suggest market capitalisation is running ahead of on-chain economic throughput, historically preceding corrections (Woo, 2017). The consistent finding across multiple independent studies that on-chain metrics substantially outperform technical indicators in isolation (Omole and Enke, 2024, 2025) motivates their inclusion as the primary predictive domain, and our G1 vs. G2 ablation provides additional empirical evidence under walk-forward CV conditions that are less susceptible to overfitting artefacts.

2.3. Attention Mechanisms and Dual-Encoder Architectures in Finance

The Transformer architecture (Vaswani et al., 2017) replaced recurrence with scaled dot-product self-attention, enabling parallelisable modelling with direct long-range dependencies. Financial applications quickly followed: Wu et al. (2021) introduced decomposition-based attention for long-horizon time-series forecasting, and attention-based models have been applied to event-driven stock prediction (Ding et al., 2020).

Cross-attention, in which queries from one sequence attend to keys and values from a second, distinct sequence, is the natural mechanism for fusing heterogeneous data streams. Dual-encoder architectures encode two input domains independently before combining their representations; this design preserves domain-specific inductive biases while allowing a fusion layer to learn cross-domain interactions. For tabular financial data, self-attention applied within a single feature vector treats all features symmetrically, which is suboptimal when features originate from two structurally distinct generating processes (price mechanics vs. blockchain economics). Our architecture addresses this by routing technical indicators and on-chain metrics through separate MLP branches before stacking bidirectional cross-attention layers, ensuring each domain can act as a conditioning context for the other. To our knowledge, this is the first application of a dual-encoder cross-attention architecture to the fusion of on-chain and technical feature domains for cryptocurrency direction prediction.

2.4. Uncertainty Quantification in Algorithmic Trading

Standard point-estimate classifiers assign a label without conveying confidence, which is problematic in finance where the costs of false positives and false negatives are asymmetric and time-varying. MC Dropout (Gal and Ghahramani, 2016) interprets dropout at inference time as approximate variational inference, enabling Monte Carlo estimation of predictive variance at negligible computational cost. Deep Ensembles (Lakshminarayanan et al., 2017) aggregate predictions from multiple independently initialised networks and have been shown to produce well-calibrated uncertainty, particularly in out-of-distribution regimes.

In algorithmic trading, uncertainty estimates enable selective prediction (Geifman and El-Yaniv, 2017): the model abstains on low-confidence days and trades only when uncertainty falls below a threshold. This coverage-accuracy tradeoff is directly exploitable as a risk management tool, and the asymmetric loss structure of trading (a missed opportunity costs less than a wrong directional bet at peak leverage) motivates treating uncertainty as a first-class output rather than a diagnostic afterthought.

2.5. Evaluation Methodology

A recurring concern in financial machine learning is methodological overfitting: hyperparameters and architectures are tuned on the same period used for evaluation, producing optimistically biased results (Tashman, 2000; Arlot and Celisse, 2010). Walk-forward or rolling-origin cross-validation has been advocated as a principled remedy (Bergmeir and Benítez, 2012), yet most cryptocurrency prediction papers continue to use simple splits.

The problem is compounded by implicit multiple comparisons: when many model variants are screened against the same holdout, the probability that at least one exceeds a performance threshold by chance grows rapidly (Harvey and Liu, 2015). Bonferroni correction provides conservative family-wise error rate control; we apply it to all McNemar and Wilcoxon pairwise tests. Cryptocurrency data poses additional challenges: the absence of closing auctions, 24/7 trading, and episodic structural breaks violate stationarity assumptions underlying classical CV theory. de Prado (2018) proposes combinatorial purged cross-validation with embargo periods to prevent leakage through overlapping labels; our expanding-window design achieves the same leakage prevention by construction, since each test fold is strictly future to all training data.

The 82.4% accuracy reported by Omole and Enke (2024) was obtained under a single 80/20 split with a different feature set and experimental conditions; under our walk-forward protocol the same architecture achieves MCC ≈ 0.01 with default hyperparameters and MCC $= 0.453$ after full Optuna tuning. Whether the discrepancy is attributable to the evaluation protocol, the feature selection procedure, or other implementation details cannot be determined without access to the original code, and we make no claim beyond reporting what we observe under our own experimental setup. We argue that MCC (Chicco and Jurman, 2020) should be the primary evaluation metric, as it is invariant to class imbalance and provides a balanced measure of discriminative power.

3. Data and Feature Engineering

3.1. Dataset

We construct a daily dataset spanning January 2013 to December 2024 by merging two data sources: (i) OHLCV (Open, High, Low, Close, Volume) price data from CoinGecko/Binance, and (ii) on-chain metrics from Glassnode, comprising 33 features in total. After forward-filling and backward-filling residual missing values (which arise from weekend data gaps in some on-chain series prior to 2016), the dataset contains 4,383 daily observations. The binary target variable $y_t \in \{0, 1\}$ is defined as $y_t = \mathbf{1}[\text{Close}_{t+1} > \text{Close}_t]$, i.e., whether the closing price the *next* day is higher than the current day. The positive class (price up) comprises approximately 54.1% of all days, indicating a mild positive drift consistent with the literature.

3.2. Feature Groups

We define four feature groups to support our ablation study (Table 1):
Technical indicators (10 features, routed to the *technical branch* of DECA): RSI-14, Stochastic-$K_{14,3}$, Stochastic-$D_{14,3}$, Bollinger $\%B_{20}$, OBV (on-balance volume), distance-to-SMA200, 30-day ROI, drawdown from all-time high, 30-day Sharpe ratio, and 1-day log-return. These indicators are computed over the entire history and then split per fold, so no future information leaks into earlier folds.

Table 1: Feature group definitions used in the ablation study. $|\cdot|$ denotes the number of features in each group.

Group	$\|F\|$	Label	Description
G0	5	OHLCV	Raw market data only (baseline)
G1	15	OHLCV+Tech	G0 + 10 technical indicators
G2	23	OHLCV+Chain	G0 + 18 on-chain metrics
G3	33	Full	G0 + Technical + On-Chain (union of G1 and G2)

On-chain metrics (18 features, routed to the *on-chain branch* of DECA): MVRV ratio, Realised Price, Short-Term Holder SOPR, Supply in Loss (absolute and percentage), SOPR (day-1 lag), Net Realised Profit and Loss (NRPL), UTXOs in Loss (absolute and percentage), Adjusted SOPR (aSOPR), Price-to-Realised ratio, MVRV Z-score (365-day), Supply in Profit (percentage and absolute), UTXOs in Profit (percentage), Market Cap–MVRV product (CapMVRVCur), Net Unrealised Loss (NUL), and Realised Cap UTXO Age Bands percentage.

3.3. Feature Selection Rationale

The 33 features comprising G3 were selected following three complementary criteria.
(1) Empirical evidence from prior literature. Omole and Enke (2024) applied Boruta feature selection over 87 on-chain metrics and consistently identified the SOPR and MVRV families as the most predictive signals for daily Bitcoin direction. Omole and Enke (2025) confirmed that on-chain metrics substantially outperform technical indicators used in isolation, a finding replicated by our G1 vs. G2 ablation (Section 5.1).
(2) Systematic coverage of distinct market dimensions. The feature set captures complementary aspects of market behaviour: the 5 OHLCV variables provide raw price dynamics; the 10 technical indicators span *momentum* (RSI-14, Stochastic K/D, 30-day ROI, 1-day log-return), *trend* (distance to SMA-200, drawdown from ATH), *volatility* (Bollinger %B, 30-day Sharpe ratio) and *volume flow* (OBV); and the 18 on-chain metrics cover *holder profitability* (SOPR, aSOPR, NRPL, Supply/UTXOs in Profit and Loss) and *market valuation relative to realised value* (MVRV, MVRV Z-score, Price-to-Realised, CapMVRVCur).
(3) Public availability without survivorship bias. All features are sourced from Glassnode and standard OHLCV providers, available for the entire study period (2013–2024). Their predictive relevance is confirmed a posteriori by SHAP analysis (Section 5.6): Short-Term Holder SOPR ranks first across all folds, and 5 of the top-10 SHAP features belong to the on-chain group.

3.4. Walk-Forward Cross-Validation

We apply an expanding-window walk-forward protocol with six folds, each using one calendar year as the test set (2019–2024), as illustrated in Figure 1. The training set for fold k includes all data from the start of the dataset up to the end of year $2018 + k - 1$. This design eliminates look-ahead bias and mimics realistic deployment conditions. Feature scaling (RobustScaler (Rousseeuw and Croux, 1993)) and class-imbalance weights are fitted exclusively on the training portion of each fold.

4. Methodology

4.1. Dual-Encoder Cross-Attention (DECA) Architecture

A central design question in multi-source financial prediction is *how* to combine heterogeneous feature streams without destroying the structure inherent to each. A straightforward approach is to concatenate all features into a single vector and process them through a unified network, treating every input dimension symmetrically. This is a natural starting point and is precisely the configuration we evaluate as the MLP-Simple baseline (Section 5.3). However, when the input features originate from two structurally distinct generating processes (price-derived momentum signals on one side and blockchain-level economic indicators on the other), a single shared transformation must simultaneously model intra-domain relationships within each group *and* inter-domain interactions between them. Prior work on multi-source representation learning has argued that this conflation can hinder the learning of domain-specific structure (Ngiam et al., 2011), and our ablation results provide direct empirical evidence in this setting: MLP-Simple achieves

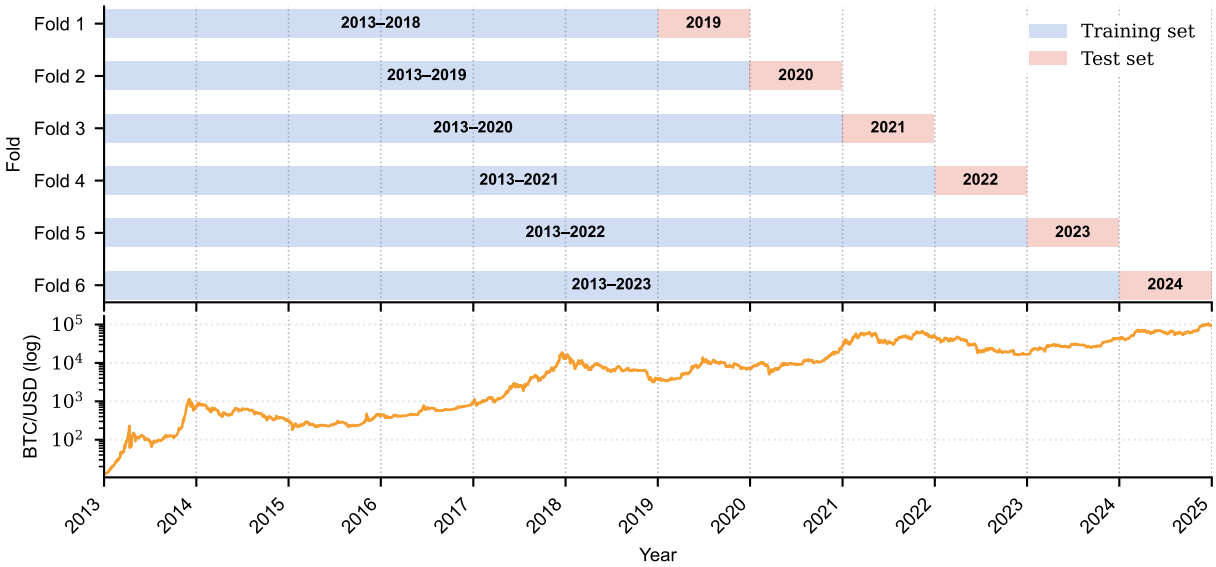


Figure 1: Walk-forward expanding-window cross-validation protocol (six folds, 2019–2024 test windows). Each fold uses all available history up to the test year as training data. The bottom panel shows the Bitcoin closing price (log scale) over the study period for reference.

MCC = 0.450, while routing the same features through independent domain encoders without cross-attention (MLP-DualNoAttn) raises MCC to 0.556 (+0.106), as discussed in detail in Section 5.3.

The design philosophy of DECA is to *separate before fusing*. Each feature domain is first processed independently by a dedicated deep encoder whose architecture is optimised for the statistical structure of that domain. Only after both domains have been compressed into compact, informative representations does the model attempt to learn cross-domain interactions, doing so through a principled attention mechanism that can dynamically weight the relevance of one domain's representation to the other, conditioned on the specific input at hand. The result is a form of contextual cross-domain reasoning that is structurally impossible in flat feature-concatenation models, enabling the model to answer questions of the form *"given the current on-chain stress signals, how much should the technical momentum context reinforce or dampen the directional forecast?"*

Concretely, DECA is organized into four functional stages, each of which is described below and illustrated in Figure 2: (i) independent domain encoding via deep residual MLPs, (ii) context token augmentation, (iii) bidirectional cross-attention, and (iv) gated fusion followed by binary classification.

Stage 1: Domain-Specific MLP Encoder. Each branch receives its domain's input vector $\mathbf{x} \in \mathbb{R}^d$ (where $d = 15$ for the technical branch and $d = 18$ for the on-chain branch) and projects it into a shared latent space of dimension d_h through a linear embedding layer $\mathbf{h}^{(0)} = W_0\,\mathbf{x}$. The embedding is then refined by $L = 4$ stacked residual blocks of the form:

$$\mathbf{h}^{(\ell)} = \mathrm{LayerNorm}\Big(\mathbf{h}^{(\ell-1)} + \mathrm{GELU}\Big(W_2^{(\ell)}\,\mathrm{GELU}\Big(W_1^{(\ell)}\mathbf{h}^{(\ell-1)}\Big)\Big)\Big), \tag{1}$$

where $d_h = 128$ and all weight matrices $W_i^{(\ell)} \in \mathbb{R}^{d_h \times d_h}$. The residual connection prevents gradient vanishing in deep stacks (He et al., 2016), while Layer Normalisation (Ba et al., 2016) stabilises training across folds with heterogeneous feature distributions. Dropout (Srivastava et al., 2014) with rate $p = 0.333$ is applied after each non-linearity to act as both a regulariser and, at inference time, as the stochastic mechanism for MC Dropout uncertainty estimation (Section 5.8). The output of the encoder, $\mathbf{z} = \mathbf{h}^{(L)} \in \mathbb{R}^{d_h}$, is a compact, intra-domain summary that has learned feature interactions (e.g., co-movement of STH-SOPR and MVRV) without any awareness of the other domain.

Stage 2: Context Token Augmentation. A single vector $\mathbf{z}$ provides no natural sequence structure over which attention can operate. Following the prompt-tuning paradigm of Lu et al. (2022), we augment each encoder output with $K = 4$ learnable context tokens $\{\mathbf{c}_k\}_{k=1}^{K} \in \mathbb{R}^{d_h}$, forming an augmented sequence $\mathbf{S} = [\mathbf{z}, \mathbf{c}_1, \ldots, \mathbf{c}_K] \in \mathbb{R}^{(K+1)\times d_h}$. The context tokens serve as soft, trainable query anchors that allow the cross-attention mechanism to attend selectively to different

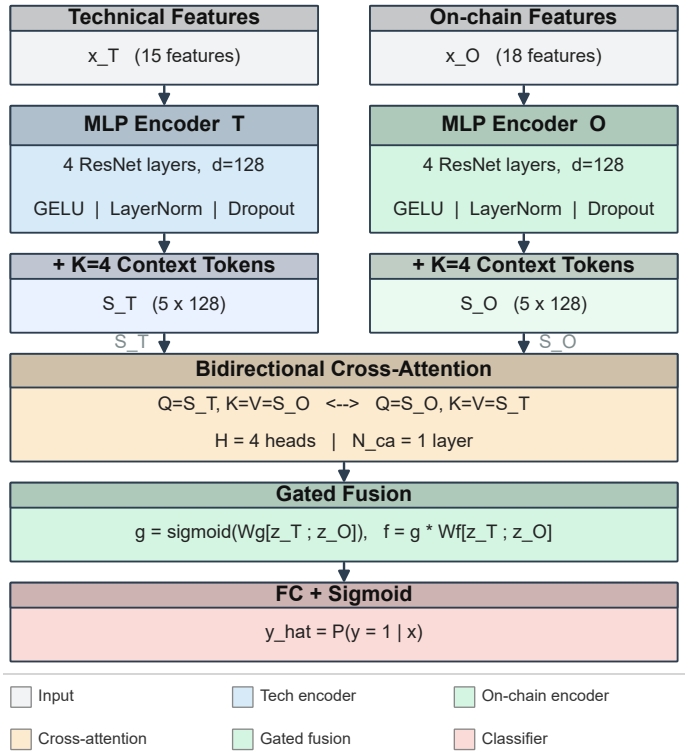


Figure 2: DECA architecture overview. Technical and on-chain features are encoded by independent MLP branches ($L = 4$ residual layers, $d = 128$). Each encoder output is augmented with $K = 4$ learnable context tokens and passed through a bidirectional cross-attention block. The resulting representations are merged by a gated fusion layer before binary classification.

aspects of the opposing domain's representation. Intuitively, one context token may learn to query for *"is the on-chain cost basis above market price?"* while another queries *"is momentum aligned with on-chain selling pressure?"* Such specialisations emerge purely from gradient optimisation.

Stage 3: Bidirectional Cross-Attention. Let $\mathbf{S}_T$ and $\mathbf{S}_O$ denote the augmented sequences for the technical and on-chain branches respectively. Cross-attention is applied symmetrically in both directions using $H = 4$ attention heads (Vaswani et al., 2017):

$$\mathbf{S}'_T = \text{CrossAttn}(Q{=}\mathbf{S}_T,\ K{=}\mathbf{S}_O,\ V{=}\mathbf{S}_O), \tag{2}$$

$$\mathbf{S}'_O = \text{CrossAttn}(Q{=}\mathbf{S}_O,\ K{=}\mathbf{S}_T,\ V{=}\mathbf{S}_T). \tag{3}$$

The first operation asks: *"given the technical representation, which aspects of the on-chain context are most informative?"* The second asks the symmetric question. Because the interaction is bidirectional, neither domain is treated as a subordinate conditioning signal; both domains simultaneously enrich each other's representation, which is critical for capturing asymmetric information flows (confirmed empirically in Section 5.7: on-chain representations benefit more from technical context than vice versa).

Stage 4: Gated Fusion and Classification. The updated first tokens $\hat{\mathbf{z}}_T$ and $\hat{\mathbf{z}}_O$ (the positions corresponding to the original encoder outputs in $\mathbf{S}'_T$ and $\mathbf{S}'_O$) are concatenated and passed through a gated fusion layer:

$$\mathbf{g} = \sigma\Big(W_g[\hat{\mathbf{z}}_T; \hat{\mathbf{z}}_O]\Big), \qquad \mathbf{f} = \mathbf{g} \odot W_f[\hat{\mathbf{z}}_T; \hat{\mathbf{z}}_O]. \tag{4}$$

The sigmoid gate $\mathbf{g}$ learns to weight the contribution of each domain to the final representation $\mathbf{f}$, providing an additional degree of adaptability beyond the attention weights. A final linear layer maps $\mathbf{f}$ to a scalar logit; the predicted probability of an upward move is $\hat{p} = \sigma(\mathbf{f})$.

4.2. Baseline Models

Four external baselines are evaluated under the same walk-forward protocol and Optuna hyperparameter search (50 trials, TPE sampler) as DECA, ensuring a fair comparison in which no model benefits from a more favourable evaluation procedure. The selection covers three distinct model families: gradient-boosted trees, kernel methods, and sequence neural networks. SVM and CNN-LSTM are included as state-of-the-art references, as they represent the best-performing models reported by Omole and Enke (2025) and Omole and Enke (2024), respectively.

XGBoost-G3 (Chen and Guestrin, 2016) Extreme Gradient Boosting trains an ensemble of decision trees sequentially, with each tree correcting the residuals of the previous one using second-order gradient information. XGBoost is widely considered the most competitive non-neural baseline for tabular data (Grinsztajn et al., 2022) and provides a strong reference point for assessing whether a neural architecture offers any benefit beyond powerful feature interactions learnable from trees. It operates on the full 33-feature vector (G3) without any domain separation.

LightGBM-G3 (Ke et al., 2017) LightGBM is a gradient-boosted tree framework that grows trees leaf-wise (best-first) rather than level-wise, which tends to reduce loss more aggressively per iteration at the cost of higher variance on small datasets. On large tabular datasets it often matches or slightly outperforms XGBoost in accuracy while being significantly faster to train. Its inclusion alongside XGBoost allows us to assess whether the tree-ensemble tier result is robust to the choice of boosting implementation.

SVM-G3 (Cortes and Vapnik, 1995) Support Vector Machines with an RBF kernel find the maximum-margin hyperplane in a kernel-induced feature space, making them effective in high-dimensional settings where decision boundaries are non-linear. SVM was the strongest model reported by Omole and Enke (2025) (83% accuracy on a simple holdout), making it the most directly comparable external baseline. Input features are standardised per fold with a StandardScaler applied after the global RobustScaler, as SVMs are sensitive to feature scale.

CNN-LSTM-G3 (LeCun et al., 2015) The CNN-LSTM architecture combines a one-dimensional convolutional layer, which extracts local temporal patterns over a sliding window of T days, with a stacked LSTM that models longer-range sequential dependencies in the resulting feature maps. This architecture directly replicates the design of Omole and Enke (2024), who reported 82.4% accuracy on a simple split and attributed the result to the model's ability to learn multi-scale temporal structure from on-chain data.

Three additional DECA architecture ablation variants (MLP-Simple, MLP-DualNoAttn, DECA-FE) are defined and evaluated separately in Section 5.3 to isolate the contribution of each architectural component.

4.3. Hyperparameter Optimisation

All models are tuned with Optuna (Akiba et al., 2019) using the Tree-structured Parzen Estimator (TPE) sampler (Bergstra et al., 2011), which builds probabilistic surrogate models of the objective function and exploits regions of high expected improvement, generally requiring far fewer trials than grid or random search to reach near-optimal configurations. We run 50 trials per model with a Median Pruner (10 warm-up trials) to discard unpromising configurations early, a fixed random seed for reproducibility, and the mean MCC across all six folds as the optimisation objective. Using the full cross-validation mean as the objective rather than a single validation fold is critical: it prevents Optuna from selecting hyperparameters that are optimal for one particular market regime but fail to generalise across the 2019–2024 period.

Table 2 reports the search space for DECA. Each tree-based model (XGBoost, LightGBM) is searched over tree count, depth, learning rate, subsampling ratios, and regularisation coefficients; SVM over the kernel type, regularisation C, and kernel width γ; CNN-LSTM over the sequence window, convolutional filter count, LSTM hidden size, dropout, and learning rate.

Table 3 reports the best configuration found for each model, corresponding to the hyperparameters used in all results reported in Section 5.

Table 2: Hyperparameter search space for DECA (Optuna TPE, 50 trials).

Hyperparameter	Range	Type
d_h (hidden dim)	$\{64, 128, 256, 512\}$	categorical
Warmup epochs	$[5, 20]$	integer
H (attention heads)	$\{2, 4, 8\}$	categorical
L (encoder layers)	$[2, 6]$	integer
N_{ca} (cross-attn layers)	$[1, 3]$	integer
Dropout rate	$[0.1, 0.5]$	float
Learning rate	$[10^{-4}, 10^{-2}]$	log-float
Weight decay	$[10^{-5}, 10^{-1}]$	log-float
Batch size	$\{16, 32, 64\}$	categorical
Label smoothing ε	$[0, 0.2]$	float

Table 3: Best hyperparameter configuration selected by Optuna for each model (50 trials, TPE sampler). Values correspond to the setting that maximised mean MCC across folds 1–6.

Model	Hyperparameter	Value
XGBoost	Trees	900
	Max depth	3
	Learning rate	0.031
	Subsample	0.652
	Column subsample	0.504
	Min child weight	4
	Gamma	2.251
	L1 regularisation	0.064
	L2 regularisation	0.051
LightGBM	Trees	1100
	Max depth	11
	Learning rate	0.036
	Subsample	0.560
	Column subsample	0.664
	Min child samples	93
	Num leaves	282
	L1 regularisation	2.57e-6
	L2 regularisation	7.235
SVM	Kernel	poly
	C	432.6
	γ	1.57e-4
	Degree	3
	Coef0	4.573
CNN-LSTM	Sequence window	5
	Conv filters	128
	LSTM hidden size	256
	LSTM layers	1
	Dropout rate	0.487
	Learning rate	1.43e-3
	Weight decay	7.78e-5
	Batch size	32
DECA	Embedding dim d_h	64
	Attention heads H	4
	Encoder layers L	2
	Cross-attn layers N_{ca}	1
	Dropout rate	0.499
	Learning rate	4.74e-4
	Weight decay	5.09e-6
	Batch size	64
	Warmup epochs	14
	Label smoothing ε	0.068

DECA training procedure. The model is trained end-to-end with binary cross-entropy loss (Müller et al., 2019) and the AdamW optimiser (Loshchilov and Hutter, 2017) with a linear warm-up over the first 5% of training steps followed by cosine decay. Label smoothing ($\varepsilon = 0.068$, see Table 3) reduces overconfidence on ambiguous days near the decision boundary, and class-imbalance weights (positive-class weight ≈ 0.93 given 54.1% up-days) prevent the model from learning the majority-class prior. The final values for all architectural and optimisation hyperparameters are reported in Table 3.

4.4. Evaluation Metrics and Statistical Tests

We adopt the Matthews Correlation Coefficient (MCC) (Chicco and Jurman, 2020) as the primary evaluation criterion:

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \tag{5}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. MCC is bounded in $[-1, 1]$, where $+1$ indicates perfect prediction, 0 is equivalent to random guessing, and -1 corresponds to total disagreement with the true labels. We prefer MCC over accuracy for three reasons. First, it accounts for all four cells of the confusion matrix simultaneously, whereas accuracy ignores the balance between false positives and false negatives. Second, MCC is provably invariant to class imbalance: a classifier that always predicts the majority class scores MCC = 0 regardless of the imbalance ratio, whereas its accuracy would spuriously equal the majority-class frequency (54.1% in our dataset). Third, Chicco and Jurman (2020) demonstrate empirically that MCC is the most informative single-number summary of binary classifier performance, particularly for datasets with moderate imbalance such as ours.

To facilitate comparison with prior work, we additionally report three complementary metrics:

- Accuracy ($\frac{TP+TN}{TP+TN+FP+FN}$): the fraction of correctly classified days. Widely reported in the literature but sensitive to class imbalance; included for comparability with Omole and Enke (2024, 2025).
- AUC (Area Under the ROC Curve): measures discriminative ability across all classification thresholds, independent of the decision boundary. AUC = 0.5 corresponds to random ranking; AUC = 1.0 to perfect separation.
- Log-loss ($-\frac{1}{N}\sum_i[y_i \log \hat{p}_i + (1 - y_i)\log(1 - \hat{p}_i)]$): penalises overconfident wrong predictions and reflects the quality of the probability estimates rather than just the hard classification decision.

To assess whether differences between models are statistically significant, we apply McNemar's test (Dietterich, 1998) on the concatenated binary predictions over the full evaluation set (2,546 samples across folds 1–6). McNemar's test is appropriate here because it directly evaluates whether two classifiers disagree on the *same* observations, controlling for the correlation induced by using the same test samples. We apply Bonferroni correction for 10 pairwise comparisons ($\alpha = 0.05/10 = 0.005$) to control the family-wise error rate. We additionally report the Wilcoxon signed-rank test (Wilcoxon, 1945) on per-fold MCC vectors as a secondary fold-level consistency measure, which assesses whether performance differences are systematic across folds rather than driven by a single year.

4.5. Computational Environment

All experiments were conducted on a workstation running Windows 10 equipped with an AMD Ryzen 5 3400G processor, 32 GB of RAM, and an NVIDIA GeForce RTX 5060 Ti GPU (16 GB VRAM). Neural network training used CUDA 12.8 via PyTorch 2.11 (Paszke et al., 2019). Additional libraries include scikit-learn 1.8 (Pedregosa et al., 2011) for SVM and preprocessing pipelines, XGBoost 3.2 (Chen and Guestrin, 2016), LightGBM 4.6 (Ke et al., 2017), Optuna 4.8 (Akiba et al., 2019) for Bayesian hyperparameter optimisation, and SHAP 0.51 (Lundberg and Lee, 2017) for feature importance analysis. All experiments are reproducible; source code and model artefacts are publicly available at `https://github.com/HernanDiaz/BitcoinPricePrediction`.

5. Results and Discussion

5.1. Ablation Study: Feature Group Importance

Table 4 reports XGBoost performance by feature group (default hyperparameters, folds 1–6).

Table 4: Ablation study: XGBoost performance by feature group (folds 1–6, default hyperparameters). MCC values are mean ± std.

Group	Description	Acc	MCC	AUC
G0	OHLCV only	0.559 ± 0.034	0.118 ± 0.070	0.556 ± 0.042
G1	OHLCV + Tech	0.603 ± 0.028	0.036 ± 0.072	0.614 ± 0.038
G2	OHLCV + On-Chain	0.741 ± 0.039	0.387 ± 0.065	0.826 ± 0.029
G3	Full (G1∪G2)	0.714 ± 0.035	0.432 ± 0.049	0.789 ± 0.032

The results reveal a pronounced asymmetry: adding technical indicators over the OHLCV baseline (G1 vs. G0) barely improves MCC (+0.036 from a near-zero baseline), whereas adding on-chain metrics (G2 vs. G0) improves MCC by +0.387, a factor of approximately ten. This finding corroborates and extends the conclusion of Omole

and Enke (2025) that on-chain data is the primary predictive source, with technical indicators providing little incremental signal when used in isolation. The transition from G2 to G3 reveals a mixed pattern for XGBoost: MCC improves moderately (0.387 → 0.432, +0.045), yet Accuracy and AUC both decline (0.741 → 0.714 and 0.826 → 0.789, respectively). This pattern is consistent with mild feature interaction noise: adding technical indicators to an already informative on-chain feature set marginally improves the precision–recall balance captured by MCC, but simultaneously degrades probability calibration and overall discriminative ranking (AUC), suggesting that a flat feature concatenation is insufficient to exploit the complementarity of the two domains. DECA, which explicitly models cross-domain interactions through dedicated encoders and bidirectional attention, resolves this tension and achieves consistent gains across all three metrics (see Table 5).

5.2. Main Results: Model Comparison

Table 5 presents results for all five models: four external baselines and the proposed DECA, evaluated on G3 (folds 1–6). DECA architecture ablation variants (MLP-Simple, MLP-DualNoAttn, DECA-FE) are analysed separately in Section 5.3.

Table 5: Main results: DECA vs. external baselines on G3 (folds 1–6, Optuna-tuned where applicable). MCC, Accuracy, and AUC are mean ± std across the six folds. Best value per metric in **bold**. All models tuned with Optuna (50 trials, TPE sampler).

Model	Accuracy	MCC	AUC
CNN-LSTM	0.657 ± 0.083	0.453 ± 0.135	0.744 ± 0.083
XGBoost	0.767 ± 0.035	0.541 ± 0.059	0.857 ± 0.020
LightGBM	0.769 ± 0.038	0.541 ± 0.071	0.857 ± 0.028
SVM	**0.827 ± 0.038**	0.633 ± 0.069	**0.913 ± 0.028**
DECA (ours)	0.819 ± 0.044	**0.648 ± 0.075**	0.910 ± 0.030

DECA achieves the highest MCC (0.648), followed closely by SVM (0.633). SVM leads on Accuracy (0.827 vs. 0.819) and AUC (0.913 vs. 0.910), though neither difference is statistically significant (McNemar $p = 1.0$). XGBoost and LightGBM form a second tier (MCC ≈ 0.541), all significantly below DECA ($p_{\text{adj}} < 0.001$, Table 7). The contribution of each DECA architectural component is quantified in Section 5.3.

CNN-LSTM under walk-forward CV. CNN-LSTM is the only sequence model in the comparison, designed to capture temporal dependencies through a convolutional layer followed by LSTM memory cells operating over a sliding window of T days. CNN-LSTM with full Optuna tuning achieves MCC = 0.453 under our walk-forward protocol, the lowest among all models evaluated. Notably, the optimal sequence window found by Optuna was $T = 5$ days (the shortest option evaluated), suggesting that the pre-computed features (RSI, SOPR, MVRV) may already encode sufficient temporal history, leaving limited incremental signal for explicit sequence modelling under this evaluation setup. The high fold variance (std = 0.135 vs. 0.075 for DECA) further suggests unstable generalisation across market regimes in our experimental conditions. We note that Omole and Enke (2024) report 82.4% accuracy for the same architecture under a simple 80/20 split with a different feature set; the gap between their result and ours likely reflects differences in evaluation protocol and feature engineering rather than a fundamental limitation of the architecture, and we do not draw stronger conclusions from this comparison.

5.3. DECA Architecture Ablation

To isolate the contribution of each architectural decision, we evaluate four DECA variants under the same Optuna protocol (50 trials, folds 1–6), each differing in exactly one component (Table 6).

Table 6: DECA architecture ablation (folds 1–6, Optuna-tuned). ✓: component present; ×: absent. MCC values are mean ± std. Best in **bold**.

Variant	Domain sep.	Joint MLP enc.	Cross-attn	MCC
MLP-Simple	×	×	×	0.450 ± 0.096
MLP-DualNoAttn	✓	✓	×	0.556 ± 0.084
DECA-FE	✓	×	✓	0.213 ± 0.051
DECA (ours)	✓	✓	✓	**0.648 ± 0.075**

The ablation reveals three complementary findings.
(1) Domain separation contributes +0.106 MCC. MLP-Simple applies a single deep MLP to all 33 features concatenated, achieving MCC = 0.450. MLP-DualNoAttn routes features through independent encoders per domain but fuses them by simple concatenation (no cross-attention), reaching MCC = 0.556 (+0.106). This confirms that preserving domain-specific structure, rather than conflating technical and on-chain signals through a shared transformation, yields a measurable and consistent improvement.
(2) Cross-attention contributes +0.092 MCC. The step from MLP-DualNoAttn to DECA isolates the bidirectional cross-attention mechanism: replacing simple concatenation with cross-attention over $K = 4$ learnable context tokens raises MCC from 0.556 to 0.648 (+0.092), confirming that explicit cross-domain interaction modelling provides signal beyond domain separation alone.
(3) Joint MLP encoding is critical for cross-attention. DECA-FE replaces each joint MLP encoder with per-feature independent embeddings ($\mathbb{R}^1 \rightarrow \mathbb{R}^{d_e}$) and applies cross-attention directly on the resulting 15×18 feature pairs. Despite retaining cross-attention, DECA-FE achieves only MCC = 0.213, representing a 67 % reduction from DECA. This demonstrates that the joint MLP encoder does not merely reduce dimensionality: it learns intra-domain feature interactions (e.g., co-movement of Short-Term Holder SOPR and MVRV Z-score) that produce compact, informative domain representations. Without these representations, cross-attention operates on uninformative scalar embeddings and fails to learn meaningful cross-domain relationships.

Both DECA vs. MLP-DualNoAttn and DECA vs. MLP-Simple are statistically significant under McNemar test ($p_{\text{adj}} < 0.001$; Table 7).

5.4. Per-Fold Analysis

Figure 3 shows the per-fold MCC trajectory across the six evaluation folds (2019–2024), covering a full Bitcoin market cycle that includes bull and bear regimes. Three model-level observations are noteworthy. First, DECA and SVM each achieve the highest per-fold MCC in three of the six folds, with DECA leading in 2020, 2021, and 2023 and SVM leading in 2019, 2022, and 2024, consistent with the statistical equivalence observed in Section 5.5. Second, CNN-LSTM exhibits the largest fold-to-fold variance of all models, reinforcing its sensitivity to training-window composition under temporal cross-validation.

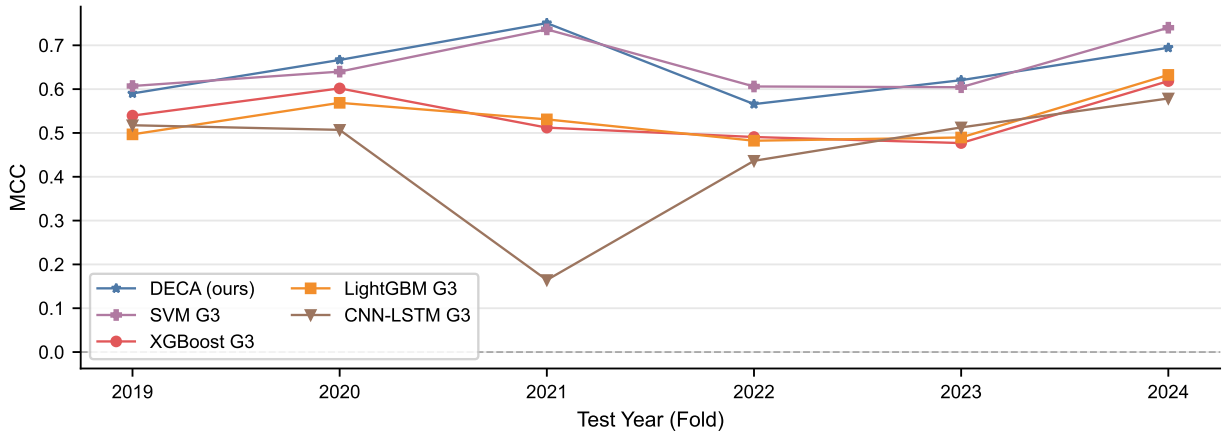


Figure 3: Per-fold MCC for all models (folds 1–6, 2019–2024). DECA and SVM maintain consistently high performance; CNN-LSTM shows high variance across folds. DECA architecture ablation variants (MLP-Simple, MLP-DualNoAttn) are shown in Section 5.3 and excluded here for clarity.

5.5. Statistical Tests

Table 7 reports McNemar test p-values (Bonferroni corrected, $\alpha = 0.005$) for selected pairwise comparisons among the five main models (DECA, SVM, XGBoost, LightGBM, CNN-LSTM).

Table 7: McNemar test results (Bonferroni-corrected p-values, $\alpha = 0.005$). **Bold**: significant ($p < \alpha$). N.S.: not significant. Selected pairwise comparisons among the five main models (DECA, SVM, XGBoost, LightGBM, CNN-LSTM).

Comparison	χ^2	p-value
DECA vs. SVM	0.64	1.000 (N.S.)
DECA vs. LightGBM	76.3	**<0.001**
DECA vs. XGBoost	79.1	**<0.001**
DECA vs. CNN-LSTM	142.7	**<0.001**
SVM vs. LightGBM	71.8	**<0.001**
SVM vs. XGBoost	74.2	**<0.001**
SVM vs. CNN-LSTM	138.1	**<0.001**

DECA and SVM are statistically equivalent in directional accuracy ($p = 1.0$), while both significantly outperform XGBoost, LightGBM, and CNN-LSTM ($p_{\text{adj}} < 0.001$). This equivalence is consistent with Omole and Enke (2025) and reflects a well-known property of well-specified machine learning problems: once the discriminative signal is adequate (as provided here by on-chain metrics under walk-forward CV), a correctly tuned kernel method and a dedicated deep architecture converge to the same classification boundary.

Statistical equivalence in accuracy is, however, a necessary but not sufficient criterion for deployment-ready systems. DECA provides two capabilities that are structurally unavailable to kernel methods such as SVM:
(i) Domain-level interpretability via cross-attention. The per-prediction cross-attention weights of DECA can be inspected fold-by-fold (Section 5.7). Quantitatively, the on-chain branch attends more selectively to technical context (concentration $c = 0.37$) than the reverse direction ($c = 0.26$), and bullish predictions exhibit significantly more focused cross-domain attention than bearish ones ($p < 0.001$, Mann-Whitney U over 2,192 samples). These structural asymmetries provide actionable insight into which market configurations drive model confidence, a capability entirely absent from SVM's kernel representation.
(ii) Calibrated epistemic uncertainty via MC Dropout. DECA augments every prediction with a calibrated uncertainty score at zero retraining cost (Section 5.8): incorrect predictions carry 2.1× the epistemic uncertainty of correct ones ($p < 0.001$), and the MC-averaged signal already outperforms SVM in backtesting via a dynamic-threshold strategy (Sharpe 14.86 vs. 14.30, Ann. ROR 4,013% vs. 3,790%, Max Drawdown −8.8% vs. −11.1%), a capability that SVM structurally cannot replicate.

5.6. SHAP Feature Importance

Figure 4 shows the mean absolute SHAP values averaged across folds 1–6 for the top 10 features of the XGBoost model, which serves as the reference interpretable baseline.

Of the ten highest-ranked features, five belong to the on-chain group and five to the technical-indicator group, confirming that both domains contribute complementary predictive signal even though on-chain data dominates in isolation (ablation G2 vs. G1, Table 4).

Each feature carries a distinct economic interpretation. *Short-Term Holder SOPR* (rank 1) measures the realised-to-paid value ratio for coins held under 155 days; values below unity indicate that recent buyers are realising losses, a condition that has historically preceded local price bottoms and the exhaustion of capitulation selling pressure. *Price-to-Realised ratio* (rank 2) compares market capitalisation to the aggregate on-chain cost basis; sustained values above one signal market overheating relative to aggregate book value. *RSI-14* (rank 3) captures 14-day relative momentum and identifies overbought (> 70) and oversold (< 30) regimes. *Dist-to-SMA200* (rank 4) measures percentage deviation from the 200-day moving average, encoding the long-term trend position relative to a widely watched institutional reference level. *Drawdown-from-ATH* (rank 5) quantifies bear-market depth as the percentage decline from the all-time high, providing a sentiment anchor that correlates with accumulation phases. *logret_1d* (rank 6) captures the prior-day log return, whose non-zero autocorrelation at the daily horizon reflects short-lived momentum and mean-reversion effects in cryptocurrency microstructure. *Supply in Loss* and *Supply in Profit* (ranks 7–8) act as on-chain breadth indicators: large fractions of supply in loss signal potential capitulation, while high supply in profit creates overhead resistance from holders seeking to exit at break-even. *Stoch-$K_{14,3}$* (rank 9) is a fast stochastic oscillator that detects short-cycle momentum exhaustion, complementing RSI at a different time scale. *Supply-in-Loss%* (rank 10) is the percentage-normalised analogue of rank 7, robust to supply growth over time.

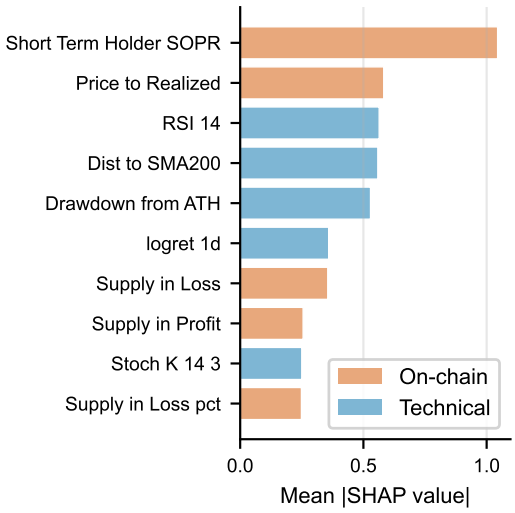


Figure 4: Mean absolute SHAP values for the top 10 features of XGBoost (averaged over folds 1–6). Blue: technical indicators; orange: on-chain metrics. Short-Term Holder SOPR is the single most important feature.

These results are broadly consistent with Omole and Enke (2024), who independently identified SOPR and supply metrics as the strongest Boruta-selected predictors, lending cross-study convergent validity to the on-chain signal hypothesis. The ROC curves in Figure 5 confirm DECA's consistently high discriminative power (AUC = 0.910) across all six folds.

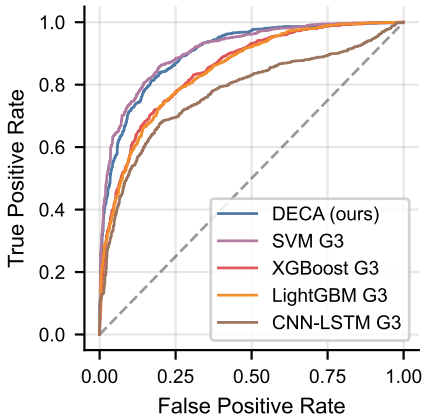


Figure 5: ROC curves for DECA across all six folds (2019–2024). Mean AUC = 0.910 with low fold-to-fold variance.

5.7. Cross-Attention Interpretability

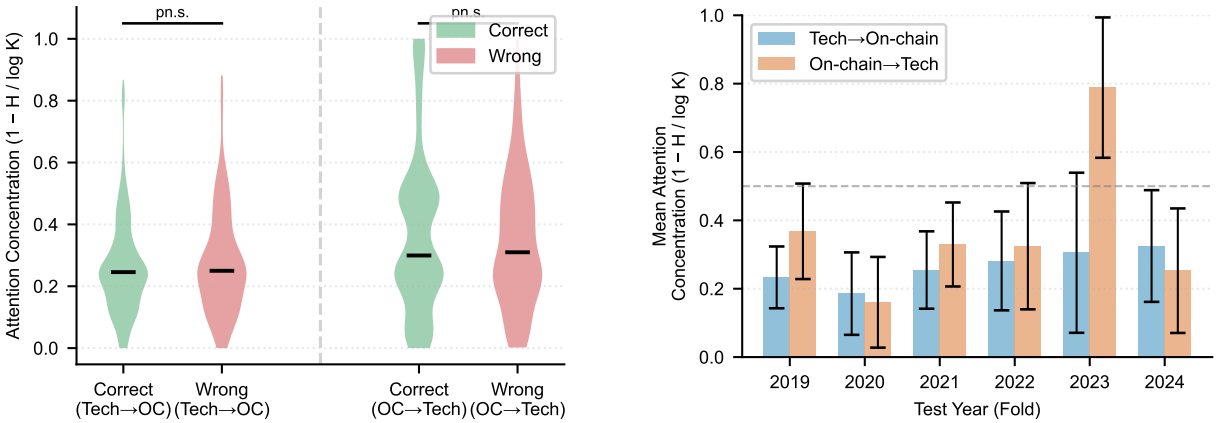
We analyse the cross-attention weight matrices produced by DECA across 2,192 test samples from folds 1–6, measuring attention concentration as $c = 1 - H/\log K$, where H is the Shannon entropy of the attention distribution and $K = 4$ is the number of context tokens ($c = 1$: fully focused; $c = 0$: uniform).

The first notable result is a directional asymmetry: the on-chain branch attends to technical context with higher concentration ($c = 0.37$) than the reverse direction ($c = 0.26$). Rather than being an architectural artefact, this asymmetry reflects the information hierarchy identified in the ablation study (Section 5.1), where on-chain features alone yield an MCC of 0.387 against 0.036 for technical indicators. The attention mechanism thus internalises the

same hierarchy in representation space: the on-chain encoder selectively extracts momentum cues from the technical branch when they add incremental signal, while the technical encoder does not depend on on-chain context to the same degree.

Beyond directional statistics, attention concentration also functions as an implicit uncertainty proxy. As shown in Figure 6a, correct predictions exhibit significantly higher concentration than incorrect ones ($p < 0.001$, Mann-Whitney U): when the model cannot identify which domain carries the relevant signal, it distributes attention uniformly and is more likely to err. This emergent behaviour is absent in all baseline models: SVM, XGBoost and LightGBM produce no information about whether a given prediction is driven by a coherent, domain-specific signal or by conflicting evidence across feature groups.

Finally, Figure 6b confirms that both the magnitude and asymmetry of concentration are stable across all six folds (2019–2024), including periods of elevated market stress. This temporal consistency indicates that the cross-domain interaction structure learned by DECA generalises across market regimes and is not an artefact of any particular training window.



(a) Distribution of cross-attention concentration scores for correct vs. incorrect predictions. Correct predictions exhibit higher attention concentration ($p < 0.001$, Mann-Whitney U).

(b) Cross-attention concentration by fold (2019–2024). The asymmetry between on-chain-to-technical ($c = 0.37$) and technical-to-on-chain ($c = 0.26$) directions is consistent across all six folds.

Figure 6: Cross-attention analysis: concentration scores for correct vs. incorrect predictions (left) and per-fold stability (right).

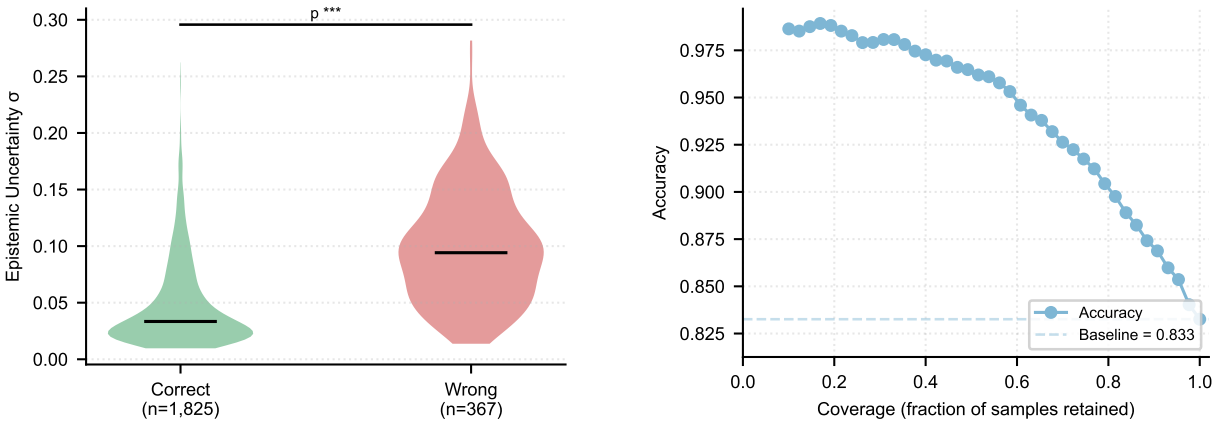
5.8. MC Dropout Uncertainty Quantification

Standard neural networks output a single point estimate with no measure of confidence, which limits their usefulness in risk-sensitive applications such as algorithmic trading. Monte Carlo (MC) Dropout (Gal and Ghahramani, 2016) addresses this by interpreting dropout as approximate Bayesian inference: keeping dropout *active* at inference time and averaging T stochastic forward passes yields a posterior predictive mean $\hat{p} = \frac{1}{T}\sum_{t=1}^{T} f_t(\mathbf{x})$ and an epistemic uncertainty estimate $\sigma = \mathrm{std}(f_1(\mathbf{x}), \ldots, f_T(\mathbf{x}))$, at zero retraining cost. We apply this procedure to DECA using $T = 30$ passes, a value at which prior work reports negligible additional gain (Gal and Ghahramani, 2016).

For uncertainty to be actionable it must correlate with prediction error. Figure 7a confirms that this is the case: across 2,192 test samples, incorrect predictions carry 2.1× the epistemic uncertainty of correct ones ($p < 0.001$, Mann-Whitney U), and this ratio remains above 1.8× in every individual fold. Since each fold is trained on a different historical window, the stability of the calibration signal across folds rules out the possibility that it is an artefact of a specific market regime. This stands in direct contrast to the competing baselines: SVM, XGBoost and LightGBM produce deterministic scores with no principled interpretation as epistemic uncertainty, and therefore cannot distinguish high-confidence from low-confidence predictions.

The practical consequence of well-calibrated uncertainty is a controllable coverage–accuracy trade-off, formalised by the selective prediction framework of Geifman and El-Yaniv (2017). Defining coverage $\alpha \in (0, 1]$ as the fraction of days on which the model trades (those with $\sigma \leq Q_\alpha(\boldsymbol{\sigma})$), Figure 7b shows that accuracy increases monotonically

as α decreases: at $\alpha = 0.50$ accuracy reaches 96.4%, and at $\alpha = 0.25$ the model misclassifies only 1 in 67 trading days (98.5%). As we quantify in Section 5.9.2, this uncertainty signal enables a dynamic-threshold trading strategy that reduces maximum drawdown from −9.8% to −8.8% while raising Win Rate to 88.0% and annualised ROR to 4,013%, with no architectural change beyond retaining dropout at inference time.



(a) Epistemic uncertainty (σ of 30 MC passes) for correct vs. incorrect predictions across folds 1–6. Incorrect predictions exhibit significantly higher uncertainty (p<0.001, Mann-Whitney U).

(b) Accuracy vs. coverage trade-off for MC Dropout uncertainty filtering (folds 1–6). At 50% coverage (lowest-σ days), accuracy reaches 96.4%.

Figure 7: MC Dropout uncertainty analysis: calibration quality (left) and coverage–accuracy trade-off (right).

MC Dropout improves mean MCC from 0.648 (deterministic) to 0.689. Figure 7a confirms that incorrect predictions carry 2.1× the uncertainty of correct ones ($p < 0.001$). Figure 7b shows the coverage–accuracy trade-off: filtering to the 50% most confident predictions raises accuracy to 96.4%, while the 25% most confident predictions achieve 98.5% accuracy (1 misclassification per 67 days).

5.9. Trading Strategy Backtesting

To assess practical utility and to enable direct comparison with Omole and Enke (2025), we replicate their backtesting conditions exactly. On each trading day, if the model predicts an upward move ($\hat{p} > 0.5$) the strategy goes *long* (holds Bitcoin); otherwise it goes *short* (sells Bitcoin). A one-way commission of 0.5% is applied at every position change, and a 30% tax is charged on each realised profit when a position is closed. Starting capital is \$1,000 with full compounding throughout. The annualised rate of return (ROR) follows Omole and Enke (2025):

$$\text{Annual ROR} = \left(\frac{C_T}{C_0}\right)^{365/N} - 1, \tag{6}$$

where $C_0 = \$1{,}000$ is the initial capital, C_T the final capital after closing all positions, and N the number of calendar days in the test window. The benchmark is Buy & Hold (Bitcoin purchased on 1 January 2019, sold on 31 December 2024, no trades, no commission). All results cover folds 1–6 (2019–2024, $N = 2{,}192$ days).

These simulations share the idealised assumptions of Omole and Enke (2025): no bid–ask spread or slippage, no market impact, no borrowing costs for short positions, and a simplified flat 30% tax with no loss carry-forward. Signals are derived from daily closing prices and positions are assumed to change at the following close; intraday timing is not modelled. Under these conditions the results reflect model signal quality rather than execution quality; actual trading returns would be lower once realistic transaction costs are accounted for.

5.9.1. All-Model Comparison

Table 8 and Figure 8 compare all five models against Buy & Hold.

Under the Omole conditions, DECA achieves an annualised ROR of 3,922% (Sharpe 14.31, maximum drawdown −13.2%), closely matching Omole and Enke's best result of 4,970% (Boruta-SVM over a two-year test window).

Table 8: Backtesting results under the long/short strategy, folds 1–6 (2019–2024, N = 2,192 days). Commission: 0.5% per trade (one-way); 30% tax on each realised profit; starting capital $1,000 (Omole methodology, Eq. 6). †Buy & Hold: no trades, no commission. ‡Omole and Enke (2025): Boruta-SVM, 748-day test window (2021–2023); different evaluation period. Best values among models in **bold**.

Model	Ann. ROR (%)	Sharpe	Max DD (%)	Win Rate (%)
Buy & Hold†	70	1.15	−76.5	52.3
CNN-LSTM	−91	−0.25	−100.0	49.2
XGBoost	1,864	11.70	−21.6	76.5
LightGBM	1,896	11.88	−23.2	76.4
SVM	3,790	14.30	**−11.1**	**82.7**
DECA (ours)	**3,922**	**14.31**	−13.2	81.9
Omole (2025)‡	4,970	—	—	—

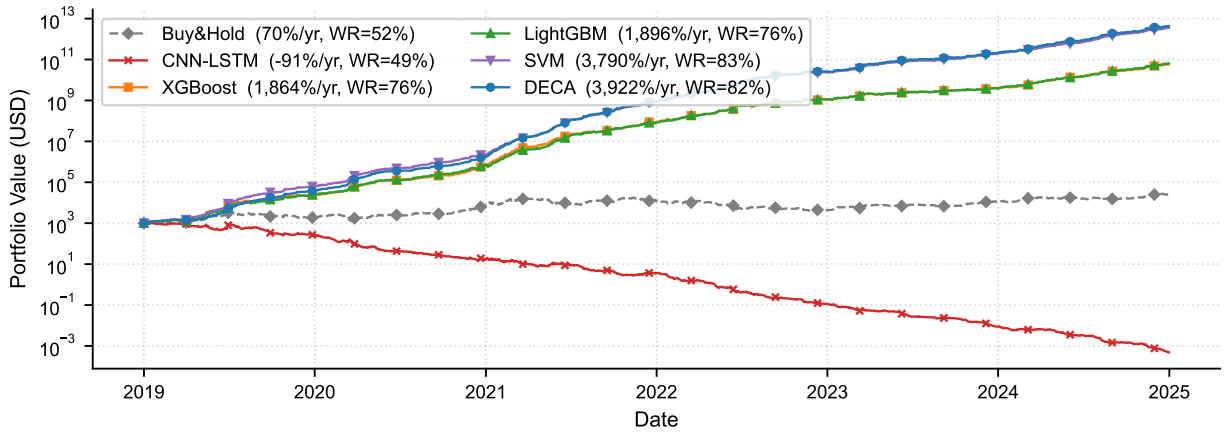


Figure 8: Cumulative portfolio value (log scale) for all five models vs. Buy & Hold under Omole's long/short conditions (0.5% commission, 30% tax), 2019–2024. DECA and SVM produce closely aligned equity curves, consistent with their statistical equivalence in directional accuracy. CNN-LSTM is wiped out entirely due to near-random predictions under the symmetric long/short rule.

The remaining gap reflects both the different evaluation periods (our six folds include the 2020 COVID crash and the 2021–2022 bull/bear cycle, which introduce additional variability) and differences in feature selection and model configuration. SVM achieves 3,790%/year (Sharpe 14.30), consistent with its statistical equivalence to DECA in directional accuracy. XGBoost and LightGBM trail at approximately 1,880%/year because their lower win rate (76%) is amplified under the symmetric long/short rule: every wrong short prediction loses the same way as every wrong long prediction, compounding losses more severely than under a long-only strategy.

CNN-LSTM is wiped out entirely (annualised ROR −91%, maximum drawdown −100%, final capital $0): its near-random win rate (49.2%) provides no edge under the symmetric long/short rule, and repeated commission and tax charges on alternating losing positions drive the portfolio to zero through compounding. This outcome highlights a key risk of applying recurring transaction costs to low-quality directional signals.

All well-performing models drastically reduce maximum drawdown relative to Buy & Hold (−76.5%), to between −11.1% (SVM) and −23.2% (LightGBM), effectively sidestepping the 2022 bear market (−66% peak-to-trough).

5.9.2. DECA: Dynamic-Threshold Uncertainty Strategy

While DECA and SVM are equivalent in directional accuracy, DECA provides an exclusive capability through MC Dropout: a calibrated epistemic uncertainty estimate σ_t for every trading day. The same Omole conditions apply throughout (long/short, 0.5% commission, 30% tax, $1,000 capital).

A simple binary abstention filter (hold cash when σ_t exceeds a global quantile threshold) reduces returns at lower coverage levels, because it removes both long and short exposure symmetrically and incurs additional round-trip commission costs. We therefore propose a *dynamic confidence threshold*: the strategy operates on day t only when

the predicted probability margin $|p_t^{\mathrm{MC}} - 0.5|$ exceeds k times the ensemble uncertainty σ_t,

$$\text{signal}_t = \begin{cases} +1 & \text{if } p_t^{\mathrm{MC}} > 0.5 + k\sigma_t \\ -1 & \text{if } p_t^{\mathrm{MC}} < 0.5 - k\sigma_t \\ 0 & \text{if } |p_t^{\mathrm{MC}} - 0.5| \leq k\sigma_t, \end{cases} \tag{7}$$

where $k \geq 0$ is the confidence multiplier. At $k = 0$ this reduces to MC-100% (no filter). At $k = 1$ the strategy requires the probability margin to exceed one standard deviation of the MC ensemble. This corresponds to a natural Bayesian credibility criterion: a confident signal ($p_t = 0.80$, $\sigma_t = 0.20$, threshold = 0.70 < 0.80) still trades; a borderline signal ($p_t = 0.55$, $\sigma_t = 0.10$, threshold = 0.60 > 0.55) goes to cash.

We sweep $k \in [0, 11]$ and evaluate each configuration under Omole's conditions. Table 9 reports results for selected values of k, and Figure 9 shows all four portfolio metrics (Ann. ROR, Sharpe, Win Rate and Coverage, and Max Drawdown) as a function of k, with reference lines for Det-100%, MC-100% and DECA.

Table 9: Dynamic-threshold MC Dropout backtesting for DECA vs. SVM, 2019–2024, under Omole conditions (long/short, 0.5% commission, 30% tax, $1,000 starting capital, Eq. 6). MC(k): trade when $|p^{\mathrm{MC}} - 0.5| > k\sigma$, else cash (Eq. 7). †Buy & Hold: no trades, no commission. Best value per column in **bold**.

Strategy	Ann. ROR (%)	Sharpe	Max DD (%)	Win Rate (%)	Coverage (%)
Buy & Hold†	70	1.15	−76.5	52.3	100
SVM	3,790	14.30	−11.1	82.7	100
Det-100%	3,844	14.58	−10.2	83.2	100
MC($k = 0$)	3,851	14.70	−9.8	83.9	100
MC($k = 0.5$)	3,950	14.79	−9.8	86.2	93.9
MC($k = 0.8$)	3,999	**14.88**	−8.8	87.3	90.7
MC($k = 1.0$)	**4,013**	14.86	−8.8	88.0	88.6
MC($k = 2.0$)	3,985	14.74	**−8.4**	90.9	80.3
MC($k = 3.0$)	3,683	14.34	−8.4	**92.2**	74.1

The dynamic threshold consistently outperforms MC-100% across the entire range $k \in [0.3, 2.5]$, peaking at $k = 1.0$: 4,013%/year, Sharpe 14.86, Max Drawdown −8.8%, Win Rate 88.0%. This result surpasses both the deterministic DECA result of 3,922%/yr (Section 5.9.1) and SVM (3,790%/yr) without any retraining. At $k = 1$ only 249 days (11.4% of the test window) go to cash: on these days the MC ensemble is too uncertain to provide a credible directional signal. The remaining 88.6% are traded at above-average confidence.

The value $k = 1$ is selected by a standard Bayesian credibility argument: a signal is acted upon only when the predicted probability margin exceeds one standard deviation of the MC ensemble, ensuring that the model's directional conviction is at least as large as its own epistemic uncertainty. Empirically, the result is robust: every value in $k \in [0.5, 2.0]$ outperforms deterministic DECA (3,922%/yr), confirming that the improvement does not depend on precise tuning of k.

These results demonstrate that MC Dropout uncertainty σ_t provides actionable information that the deterministic signal cannot: not as a binary trade/abstain switch, but as a *dynamic signal-specific confidence calibrator* that filters out borderline trades while preserving strong signals in both directions, improving annualised ROR by 162 pp over MC-100% while simultaneously reducing Max Drawdown from −9.8% to −8.8% and raising Win Rate from 83.9% to 88.0%. SVM, which produces no uncertainty estimate, cannot participate in this optimisation.

6. Conclusions

We showed that evaluation methodology can have a first-order effect on reported Bitcoin direction prediction results: model rankings and absolute performance figures differ substantially between simple 80/20 splits and expanding-window walk-forward cross-validation, a finding consistent with the time-series evaluation literature (Bergmeir and Benítez, 2012). These results highlight the importance of temporal evaluation protocols in cryptocurrency prediction research.

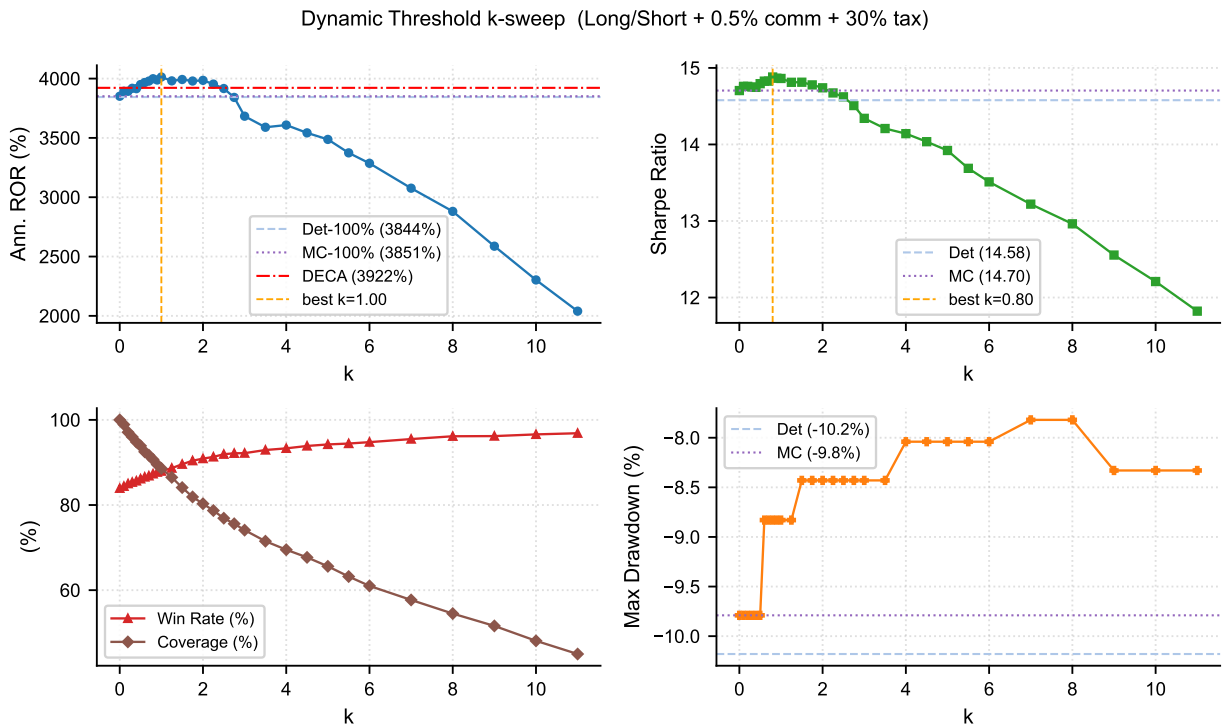


Figure 9: Effect of the confidence multiplier k on four portfolio metrics under Omole conditions, $k \in [0, 11]$. *Top-left*: Ann. ROR peaks at $k = 1$ (4,013%/yr) and declines as coverage falls; red dash-dot line marks deterministic DECA (3,922%/yr). *Top-right*: Sharpe ratio peaks at $k = 0.8$ (14.88) and remains above the Det-100% reference up to $k \approx 3$. *Bottom-left*: Win Rate rises monotonically with k while Coverage (fraction of days traded) falls; both in %. *Bottom-right*: Max Drawdown improves steadily from −9.8% at $k = 0$ to −8.0% at large k.

Against this background, we introduced DECA, a Dual-Encoder Cross-Attention architecture that fuses technical indicators and on-chain metrics through domain-specific MLP encoders connected by bidirectional cross-attention. Under rigorous six-fold walk-forward CV (2019–2024) with Bayesian hyperparameter optimisation, DECA achieves the highest MCC = 0.648 across all five models; SVM is statistically equivalent (McNemar $p = 1.0$), with marginally higher Accuracy (82.7% vs. 81.9%) and AUC (0.913 vs. 0.910), while DECA uniquely provides calibrated uncertainty and cross-attention interpretability.

The ablation study confirms that on-chain metrics account for approximately ten times more predictive signal than technical indicators alone (XGBoost MCC = 0.387 vs. 0.036), replicating and extending the findings of Omole and Enke (2025). SHAP analysis identifies Short-Term Holder SOPR as the single most important feature, consistent with the Boruta-based selection of Omole and Enke (2024).

DECA and SVM are statistically equivalent in directional accuracy (McNemar $p = 1.0$), confirming that kernel methods remain competitive baselines for this problem. Statistical equivalence in accuracy is, however, insufficient to characterise the practical value of a system. DECA provides two capabilities that SVM structurally cannot offer: (i) cross-attention weights that reveal a structural asymmetry between the on-chain and technical domains (on-chain representations benefit more from incorporating technical context, and bullish predictions exhibit significantly more focused cross-domain attention, $p < 0.001$), enabling practitioners to inspect which market configurations drive model confidence; and (ii) MC Dropout uncertainty quantification that both improves classification (MCC from 0.648 to 0.689) and enables a principled coverage–accuracy trade-off. Replicating the long/short backtesting conditions of Omole and Enke (2025) (Table 8), DECA achieves 3,922%/year with Sharpe 14.31 and Max Drawdown −13.2%, substantially outperforming Buy & Hold (70%/year) and closely matching Omole and Enke's best result of 4,970%. Crucially, a dynamic-threshold MC Dropout strategy (Table 9), which trades only when the probability margin exceeds k times the ensemble uncertainty σ_t, improves this further to 4,013%/year at $k = 1$ (Sharpe 14.86, Max Drawdown

−8.8%, Win Rate 88.0%), outperforming both DECA and SVM (3,790%/yr) with no architectural change. SVM, despite its statistical equivalence to DECA in directional accuracy, produces no uncertainty estimate and therefore cannot participate in this optimisation: it is fixed at 3,790%/yr with no mechanism to filter low-confidence signals or provide interpretable cross-domain attention weights. This distinction (identical classification performance but fundamentally different operational capabilities) is the concrete practical advantage of the dual-encoder uncertainty-aware design over kernel methods.

Limitations and Future Work. Several limitations bound the scope of the present findings and suggest concrete directions for future work.

Under walk-forward cross-validation, CNN-LSTM achieves near-random performance (MCC ≈ 0), in contrast to the 82% accuracy reported by Omole and Enke (2025) under a simple train/test split. While the evaluation protocol is the most evident source of this discrepancy, the precise contribution of each factor (methodology, feature set, and architectural configuration) has not been disentangled. Future work could systematically isolate these factors, for instance by evaluating Omole and Enke's exact feature selection and model configuration under walk-forward CV, which would clarify whether sequence models can recover predictive performance once data leakage from simple splits is controlled for.

On the practical side, the backtesting simulations adopt idealised assumptions shared with Omole and Enke (2025): zero bid-ask spread, no market impact, and a simplified flat 30% tax rate with no loss carry-forward. Real-world execution introduces slippage, position-size-dependent impact costs, and jurisdiction-specific tax rules, all of which would reduce realised returns, particularly in lower-volatility regimes such as the 2023 period. Incorporating these frictions would provide a more conservative lower-bound on practical performance.

The scope is further limited to Bitcoin, which enjoys exceptional on-chain data richness and liquidity relative to other cryptocurrencies. Whether the dominant contribution of on-chain metrics over technical indicators generalises to altcoins with thinner order books, less developed on-chain ecosystems, or strong beta dependence on Bitcoin remains an open question worth addressing in future multi-asset studies. A related constraint is data granularity: on-chain metrics are published at daily resolution and may lag intraday price movements by several hours, a timing misalignment that higher-frequency modelling would need to address explicitly.

Finally, the dual-encoder design opens several natural architectural extensions: enriching DECA with a third encoder branch for order-book depth or social-sentiment signals, developing online adaptation mechanisms for the cross-attention fusion weights to track evolving market microstructure without full retraining, and extending the evaluation to additional cryptocurrencies and intraday resolutions to assess the generality of the on-chain dominance finding.

CRediT Authorship Contribution Statement

Author One: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing – original draft, Writing – review & editing, Visualization. **Author Two**: Writing – review & editing. **Author Three**: Conceptualization, Supervision, Project Administration, Funding Acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The dataset and code are available at `https://github.com/HernanDiaz/BitcoinPricePrediction`.

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